

Data Vortex — Round 1 Phase 1

Exploratory Data Analysis Report

1. Dataset Overview

The recovered dataset consisted of 12,360 raw post records and 1,500 user records. After removing 360 exact duplicate rows, the cleaned posts dataset contains 12,000 rows. The users table required no corrections: no nulls, no duplicate user IDs, and full referential integrity with the posts table (every user_id in posts exists in users).

2. Cleaning Summary

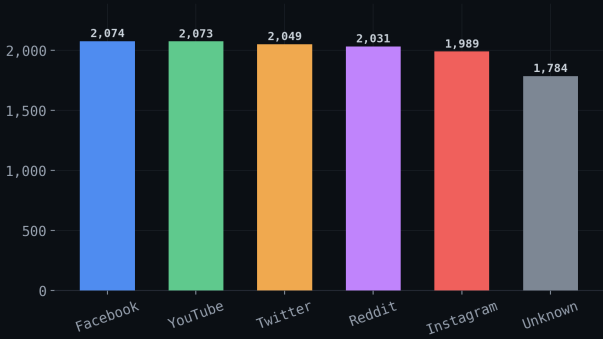
Issue	Column	Count	Fix Applied
Exact duplicate rows	post_id	360	Dropped, kept first occurrence
Mixed timestamp formats (epoch / DD-MM-YYYY / ISO8601)	timestamp	12,360	Parsed to single datetime dtype
Missing platform	platform	1,784	Filled as "Unknown" (no basis to infer)
Literal "NULL" strings	text_content	23	Converted to true NaN
HTML-escaped text	text_content	328	Unescaped (& → &, etc.)
Genuinely missing text	text_content	1,711	Left as NaN, not fabricated
Negative likes (impossible)	likes	509	Converted to NaN before imputation
Missing/negative likes	likes	2,323	Imputed with platform-median, flagged in likes_imputed

3. Visual Summary

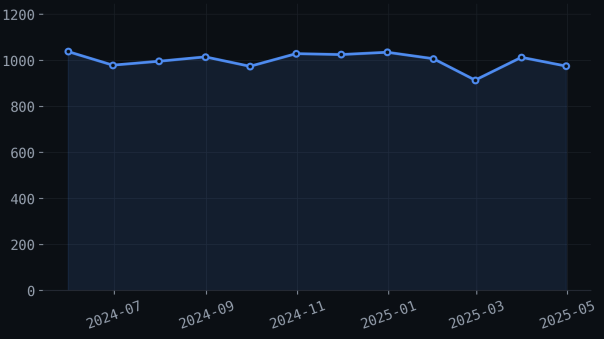
DATA VORTEX — PHASE 1 EDA

social_engine_posts_cleaned.csv · 12,000 rows

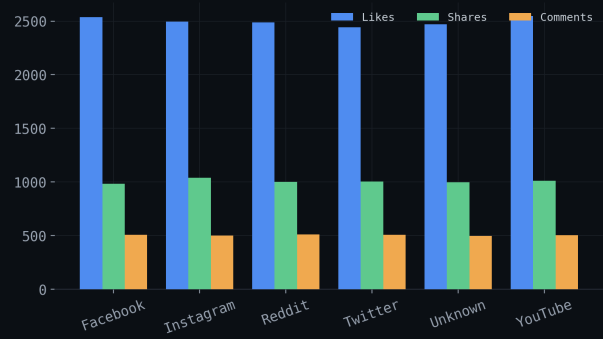
POST VOLUME BY PLATFORM



POSTS PER MONTH



AVERAGE ENGAGEMENT BY PLATFORM



USERS BY LANGUAGE



4. Key Findings

Platform distribution. Roughly even across the 5 real platforms (1,989–2,074 posts each), with 1,784 posts of unknown platform. No single platform dominates the corruption, suggesting it was applied close to uniformly at random rather than targeting one platform.

Posting volume over time. Stable at roughly 975–1,040 posts/month from May 2024 to April 2025, with no visible growth, decay, or seasonal spike.

Engagement correlation. Likes, shares, comments, and follower_count show near-zero correlation with each other ($|r| \leq 0.02$). This indicates the engagement numbers in this dataset are synthetic or randomly generated rather than reflecting real social dynamics — a more popular account (higher followers) does not get more likes here. This is an important caveat for Phase 2: a SQL correlation analysis on this data correctly returning near-zero correlation is the right answer, not a bug in the query.

User base diversity. Spans 10 languages fairly evenly (1,119–1,369 users each) and is geographically diverse, with no single location dominating.

5. Files

- cleaned_posts.csv / cleaned_users.csv — cleaned output tables
- cleaned_dataset_merged.csv — single-file join of posts and users on user_id, for submission
- clean_pipeline.py — full reproducible cleaning script
- cleaning_log.txt — plain-text log of every transformation applied
- eda_charts.png — platform volume, posting trend, engagement, and language distribution charts